

Solution STA510 Exam fall 2024

Problem 1

Assume that $X \sim \text{Pareto}(a, b)$ distributed with location parameter $a > 0$ and shape parameter $b > 0$. The density function $f(x)$ of X is given by

$$f(x) = \frac{ba^b}{x^{b+1}}, \quad \text{for } 0 < a < x \text{ and } 0 < b.$$

a)

Show that $f(x)$ is a proper density function.

For $f(x)$ to be a proper density, we must have that $f(x) \geq 0$ which is clearly the case as $0 < a < x$ and $b > 0$, i.e. all elements are positive. In addition,

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_a^{\infty} \frac{ba^b}{x^{b+1}} dx = a^b \int_a^{\infty} bx^{-b-1} dx \\ &= a^b \cdot [-x^{-b}]_a^{\infty} = -a^b \cdot (0 - (a^{-b})) = a^b \cdot a^{-b} = 1 \end{aligned}$$

and thus $f(x)$ is a proper density function.

b)

Show that the cumulative distribution function $F(x) = 1 - \left(\frac{a}{x}\right)^b$ for $0 < a < x$ and $0 < b$.

$$\begin{aligned} F(x) &= P(X \leq x) = \int_{-\infty}^x f(y) dy = \int_a^x \frac{ba^b}{y^{b+1}} dy \\ &= a^b \cdot [-y^{-b}]_a^x = -a^b \cdot (x^{-b} - (a^{-b})) = -\frac{a^b}{x^b} + 1 \\ &= 1 - \left(\frac{a}{x}\right)^b. \end{aligned}$$

c)

Explain (with words and formulas, not code) how we can simulate $n = 10000$ random variables from this distribution using the inverse transformation method.

We know that if X is a continuous random variable with cdf $F(x)$ then $U = F(X) \sim U[0, 1]$. Thus, we solve the equation

$$\begin{aligned} 1 - \frac{a^b}{X^b} &= U \iff -\frac{a^b}{X^b} = U - 1 \iff \frac{a^b}{X^b} = 1 - U \\ &\iff X^b = \frac{a^b}{1 - U} \iff X = \frac{a}{(1 - U)^{1/b}}. \end{aligned}$$

Now, we generate $U_i \sim U[0, 1]$ and get that $a(1 - U)^{-1/b} \sim \text{Pareto}(a, b)$ for $i = 1, \dots, 10000$.

d)

```

r.pareto1 <- function(n){
  return(4/sqrt(runif(n,2,4)))
}
r.pareto2 <- function(n){
  return(4/sqrt(runif(n)))
}
r.pareto3 <- function(n){
  return(4/sqrt(1-runif(n)))
}
r.pareto4 <- function(n){
  return(2/(1-runif(n)^(1/4)))
}

```

Given $a = 4$ and $b = 2$, which of the above functions correctly simulates random numbers with density $f(x)$? Argue why the functions produce wrong or correct results.

Given $a = 4$ and $b = 2$, the inverse cumulative distribution function is $F^{-1}(U) = \frac{4}{\sqrt{1-U}}$.

The first function `r.pareto1` is not correct as we are sampling r.v. from a $U(2, 4)$ distribution instead from a $U(0, 1)$ distribution.

The third function `r.pareto3` is correct as $F^{-1}(U)$ is as specified above, i.e. $F^{-1}(U) = \frac{4}{\sqrt{1-U}}$ and we are sampling from a $U(0, 1)$ distribution.

The second function `r.pareto2` is also correct. The function is identical to `r.pareto3` except that we have $\sqrt{U}$ instead of $\sqrt{1-U}$. However, if $U \sim U(0, 1)$, then also $(1-U) \sim U(0, 1)$.

The fourth function `r.pareto4` is defined as $\frac{2}{(1-U)^{1/4}}$ which is not equal to $F^{-1}(U)$ and is therefore not correct.

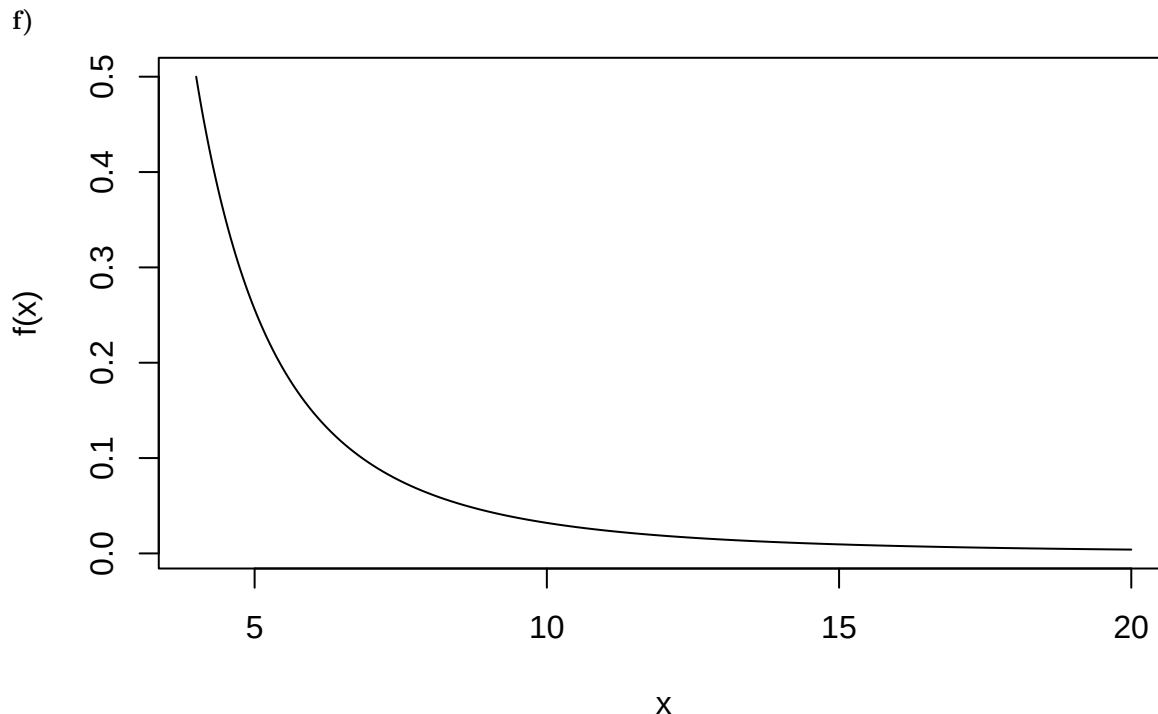
e)

Given $a = 4$ and $b = 2$, calculate $P(X < 6)$ and $P(10 < X < 15)$.

Given $a = 4$ and $b = 2$, we have $F(x) = 1 - \left(\frac{4}{x}\right)^2$ and

$$P(X < 6) = F(6) = 1 - \left(\frac{4}{6}\right)^2 = \frac{5}{9} = 0.556,$$

$$P(10 < X < 15) = F(15) - F(10) = -\left(\frac{4}{15}\right)^2 + \left(\frac{4}{10}\right)^2 = \frac{4}{45} = 0.889.$$



Given $a = 4$ and $b = 2$, the density $f(x)$ is shown in the above figure. Assume we have simulated $n = 10000$ random variables and we want to create a histogram. Would you recommend to use the “Scott” rule for the automatic bin-width selection? Why?

The Scott rule assumes that the observed data is normal distributed. However, the $Pareto(a = 4, b = 2)$ density deviates significantly from a normal distribution. Thus it might be wise to choose a different method. The Scott rule is designed to minimize the mean integrated squared error between the histogram and an underlying normal distribution.

g)

Assume now we want to estimate the integral

$$I = \int_a^\infty \frac{\ln(x)}{x^b} dx, \quad \text{for } a > 0, b > 0.$$

Explain how we can estimate I by a generalized crude Monte Carlo estimator and write down an expression for the Monte Carlo estimator based on n random draws from a $Pareto(a, b)$ distribution.

Let $g(x) = \frac{\ln(x)}{x^b}$, then we can decompose $g(x)$ as $g(x) = h(x)f(x)$ with $f(x) = \frac{ba^b}{x^{b+1}}$ and $h(x) = \frac{x \ln(x)}{ba^b}$ and

$$I = \int_a^\infty h(x)f(x)dx = E(h(X))$$

where $X \sim Pareto(a, b)$ and

$$\hat{I}_{GCMC} = \frac{1}{n} \sum_{i=1}^n h(X_i) = \frac{1}{nba^b} \sum_{i=1}^n X_i \ln(X_i)$$

where $X_1, \dots, X_n \stackrel{iid}{\sim} Pareto(a, b)$.

h)

Explain if and how we can estimate I by an antithetic Monte Carlo estimator. Write down an expression for the antithetic Monte Carlo estimator based on n random draws from a *Pareto*(a, b) distribution.

For the antithetic Monte Carlo estimator, we need $g(x) = \frac{\ln(x)}{x^b}$ to be monotonic on the interval (a, ∞) . This depends on the choice of the parameters a and b . If $g(x)$ is monotonic on (a, ∞) , then

$$\hat{I}_{GAT} = \frac{1}{n} \sum_{i=1}^{n/2} h(X_i) + h(Y_i) = \frac{1}{nba^b} \sum_{i=1}^{n/2} X_i \ln(X_i) + Y_i \ln(Y_i)$$

with $X_i = F^{-1}(U_i) = \frac{a}{(1-U_i)^{1/b}}$ and $Y_i = F^{-1}(1 - U_i) = \frac{a}{(U_i)^{1/b}}$ and $U_1, \dots, U_{n/2} \sim U(0, 1)$.

Problem 2

Assume we can only generate random numbers from the standard uniform distribution $U(0, 1)$ and the standard normal distribution $N(0, 1)$.

a)

Describe how we can generate random numbers X which are Lognormal distributed with mean $\mu = 1$ and variance $\sigma^2 = \frac{1}{4}$.

We have, if $W_1 \sim N(0, 1)$, then $W_2 = \mu + \sigma W_1 \sim N(\mu, \sigma^2)$, and $e^{W_2} \sim \text{Lognormal}(\mu, \sigma^2)$. Thus,

$$X = e^{\mu + \sigma W_1} = e^{1 + \frac{1}{2} W_1} \sim \text{Lognormal}(\mu = 1, \sigma^2 = \frac{1}{4}).$$

b)

Describe how we can generate random numbers Y which are χ^2 -distributed with 2 degrees of freedom.

The easiest way is to assume $W_1, W_2 \sim N(0, 1)$, then we know that

$$Y = W_1^2 + W_2^2 \sim \chi_2^2.$$

Alternatively, we could assume $U \sim U(0, 1)$, then we have

$$Y = -\frac{2\beta \ln(U)}{\beta} = -2\ln(U) \sim \chi_2^2.$$

c)

Let $Z = 5 \cdot X + 3 \cdot Y$. Describe how we can find $E(Z)$, $P(Z < 20)$ and $P(40 < Z < 60)$ by $n = 10000$ Monte Carlo simulations.

The random variable Z is defined by a linear combination of X and Y , namely $Z = 5 \cdot X + 3 \cdot Y$. Thus we can find the expectation and the probabilities by Monte Carlo simulations.

1. Draw $n = 10000$ random numbers by $X_1, \dots, X_n \sim \text{Lognormal}(\mu = 1, \sigma^2 = \frac{1}{4})$ as described in a).
2. Draw $n = 10000$ random numbers by $Y_1, \dots, Y_n \sim \chi_2^2(\mu = 1, \sigma^2 = \frac{1}{4})$ as described in b).
3. Then calculate $Z_i = 5 \cdot X_i + 3 \cdot Y_i$, $i = 1, \dots, n$.

The MC estimator for $E(Z)$ is

$$\bar{Z} = \widehat{E(Z)} = \frac{1}{n} \sum_{i=1}^n Z_i.$$

and $p_1 = P(Z < 20)$ and $p_2 = P(40 < Z < 60)$ are

$$\hat{p}_1 = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(Z_i < 20),$$

$$\hat{p}_2 = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(40 < Z_i < 60),$$

where $\mathbb{I}$ is the indicator function with $\mathbb{I}(a < x < b) = 1$ if $a < x < b$ and $\mathbb{I}(a < x < b) = 0$ otherwise.

d)

Assume we want to estimate the integral

$$I = \int_1^3 \int_{-2}^2 \int_0^\infty \frac{x}{y^2 + 2z} e^{-\frac{1}{4}x^2} dx dy dz.$$

Write down an expression for a Monte Carlo estimator of I based on n random draws from a uniform and/or normal distribution.

We can write this integral as the product of two integrals, i.e.

$$I = \int_1^3 \int_{-2}^2 \int_0^\infty \frac{x}{y^2 + 2z} e^{-\frac{1}{4}x^2} dx dy dz = \int_0^\infty x e^{-\frac{1}{4}x^2} dx \cdot \int_1^3 \int_{-2}^2 \frac{1}{y^2 + 2z} dy dz = I_1 \cdot I_2.$$

with $I_1 = \int_0^\infty x e^{-\frac{1}{4}x^2} dx$ and $I_2 = \int_1^3 \int_{-2}^2 \frac{1}{y^2 + 2z} dy dz$.

The first integral can be written as

$$I_1 = \int_0^\infty (\sqrt{2\pi}\sigma x) \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{(x-0)^2}{\sigma^2}} dx = \int_{-\infty}^\infty \mathbb{I}(x \geq 0) \sqrt{2\pi}\sigma x f(x) dx = \int_{-\infty}^\infty h(x) f(x) dx$$

with $f(x)$ the density function of a normal distribution with $\mu = 0$ and $\sigma^2 = 2$, and $h(x) = \mathbb{I}(x \geq 0) \sqrt{2\pi}\sigma x = \mathbb{I}(x \geq 0) 2\sqrt{\pi}x$. Then I_1 can be estimate by $X_i \sim N(\mu = 0, \sigma^2 = 2)$ and

$$\hat{I}_1 = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(x \geq 0) 2\sqrt{\pi} X_i.$$

The second integral I_2 , we can estimate by uniform distributed variables, namely $Y_i \sim U[-2, 2]$ and $Z_i \sim U[1, 3]$ resulting into

$$\hat{I}_2 = \frac{(2 - (-2))(3 - 1)}{n} \sum_{i=1}^n \frac{1}{Y_i^2 + 2Z_i} = \frac{8}{n} \sum_{i=1}^n \frac{1}{Y_i^2 + 2Z_i}.$$

Thus the final estimate is

$$\hat{I} = \hat{I}_1 \cdot \hat{I}_2 = \frac{1}{n} \left(8 \sum_{i=1}^n \frac{1}{Y_i^2 + 2Z_i} \cdot \sum_{i=1}^n \mathbb{I}(x \geq 0) 2\sqrt{\pi} X_i \right).$$

Problem 3

Suppose that X is a random variable which describes the number of newborns in a given time unit.

a)

Explain why its natural to model X by Poisson distribution with density function $p(x|\mu)$.

We are interested in the number of times an event (childbirth) occur inside a time interval. This is the typical set up for a Poisson distribution. The situation describes a counting process $N(t)$ with intensity parameter λ or $\lambda(t)$ where we have $N(0) = 0$, independent increments, $P(N(s + \Delta s) - N(s) = 1) \approx \lambda \Delta s$ or $\lambda(t) \Delta s$ and $P(N(s + \Delta s) - N(s) > 1) \approx 0$.

b)

Assume the expected number of newborns at the Stavanger University hospital is 6 per day and the arrival of newborns can be modeled by a homogeneous Poisson process.

What is the probability that we have less than 3 newborns at a given day?

$$\begin{aligned} P(X < 3) &= \sum_{i=0}^2 P(X = i) = \sum_{i=0}^2 f(x_i) = \sum_{i=0}^2 \frac{\mu^{x_i}}{x_i!} e^{-\mu} \\ &= e^{-6} \sum_{i=0}^2 \frac{6^{x_i}}{x_i!} = e^{-6} \left(\frac{6^0}{0!} + \frac{6^1}{1!} + \frac{6^2}{2!} \right) \\ &= e^{-6} (1 + 6 + 18) = 25e^{-6} = 0.06. \end{aligned}$$

There is a probability of 6% that we have less than 3 newborns at a given day.

c)

What is the probability that it takes more than 8 hours until the third baby is born on a given day

We can define 3 events T_1 ="time until baby number 1", T_2 ="time between birth of baby number 1 and baby number 2" and T_3 ="time between birth of baby number 2 and baby number 3". We know the times T_i between events is exponential distributed in a Poisson process. Further, we can define a r.v. $S = T_1 + T_2 + T_3$ which describes the time until the third birth on a given day. We know that $S \sim \text{Gamma}(\alpha = 3, \beta = 1/6)$ ($\beta = 1/\lambda$ and $\lambda = 6$). Given $\Gamma(\alpha) = \Gamma(3) = 2! = 2$ and $\int x^2 e^{ax} dx = e^{ax} \left(\frac{x^2}{a} - \frac{2x}{a^2} + \frac{2}{a^3} \right) + C$ (from the formula sheet), we have

$$\begin{aligned} P(S > 8\text{hours}) &= P(S > 1/3\text{days}) = \int_{1/3}^{\infty} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} dx \\ &= \int_{1/3}^{\infty} \frac{1}{(1/6)^3 \cdot 2} x^{3-1} e^{-6x} dx \\ &= \frac{6^3}{2} \int_{1/3}^{\infty} x^2 e^{-6x} dx \\ &= \frac{6^3}{2} \left[e^{-6x} \left(\frac{x^2}{-6} - \frac{2x}{(-6)^2} + \frac{2}{(-6)^3} \right) \right]_{1/3}^{\infty} \\ &= \frac{6^3}{2} \left(0 - e^{-6/3} \left(-\frac{(1/3)^2}{6} - \frac{2/3}{(6)^2} - \frac{2}{(6)^3} \right) \right) \\ &= \frac{6^3}{2} \left(e^{-2} \left(\frac{1}{9 \cdot 6} + \frac{2}{3 \cdot 36} + \frac{2}{6^3} \right) \right) = 0.677 \end{aligned}$$

There is a chance of 67.7% that it takes more than 8 hours until the third baby is born on a given day.

Alternatively, we can re-formulate the problem as the probability of having less than 3 babies in 8 hours with $\lambda t = 6 \cdot 1/3 = 2$ and

$$P(S > 8\text{hours}) = P(X < 3) = e^{-2} \left(\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} \right) = (1 + 2 + 2)e^{-2} = 0.677$$

d)

Describe one possibility how we can simulate times of birth for 30 days.

In the lecture we have learnt two options to simulate a homogeneous Poisson process.

Option 1:

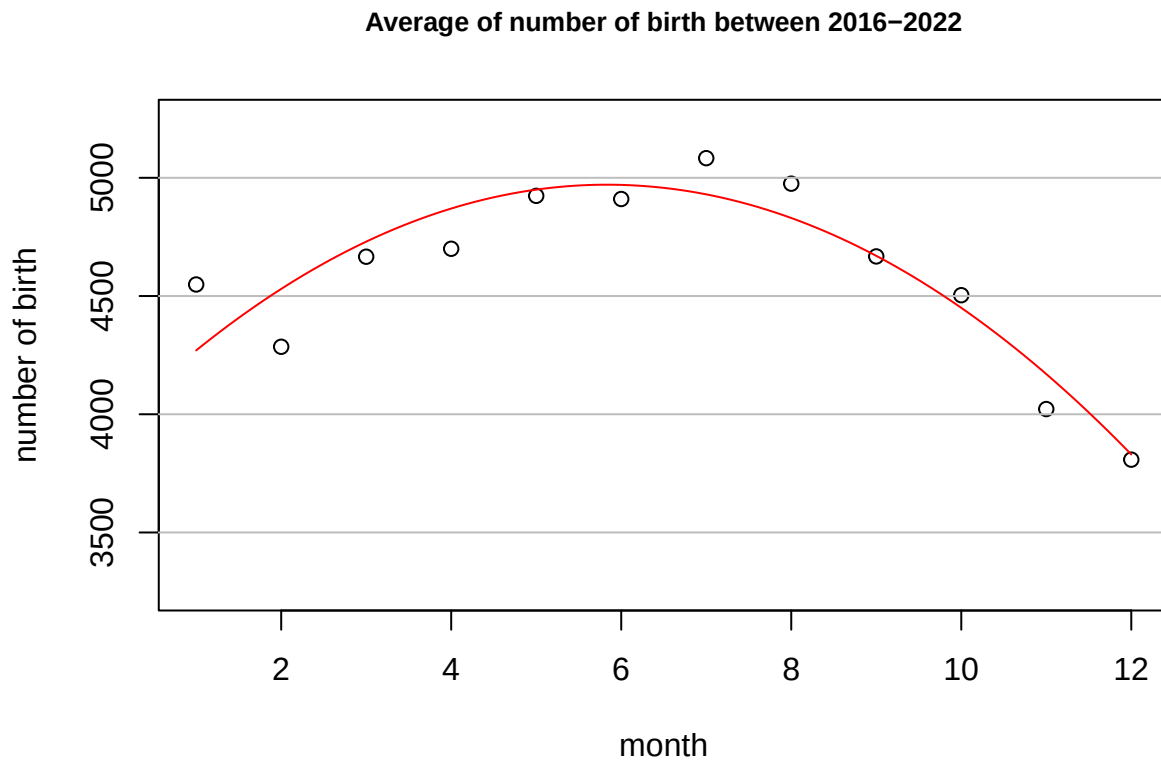
Simulate interarrival times $T_1, T_2, \dots$ from the exponential distribution with parameter $\lambda = 6$. Stop simulating interarrival times if $\sum_{i=1}^n T_i \leq 30$ and $\sum_{i=1}^{n+1} T_i > 30$ and keeping the interarrival times $T_1, \dots, T_n$.

Option 2:

First simulate the number of events, n , in an interval $[0, 30]$ from the Poisson distribution with expectation $\lambda t = 6 \cdot 30 = 180$. Then simulate the arrival times inside this interval by simulating n random numbers from the Uniform $[0, 30]$ distribution and sort the numbers.

e)

The number of newborns in Norway in a year can be modeled by a nonhomogeneous Poisson process with intensity $\lambda(t) = -30t^2 + 350t + 3950$. The below figure visualizes the average number of birth per months between 2016-2022 (small black circles) and the fitted intensity $\lambda(t)$ (red solid line).



What is the expected value of newborns in Norway for a year?

The expected number of newborns in Norway for a year is

$$\begin{aligned}\int_0^{12} \lambda(t) dt &= \int_0^{12} -30t^2 + 350t + 3950 dt \\ &= \left[-30 \frac{1}{3} t^3 + 350 \frac{1}{2} t^2 + 3950t \right]_0^{12} = 55320.\end{aligned}$$

f)

Describe how we can simulate times of birth for 30 days by a non-homogeneous Poisson process with intensity $\lambda(t)$ (visualized by the red line in the above figure). Specify potential constants.

From the figure we see that $\lambda(t) = -30t^2 + 350t + 3950$ is bounded, i.e. $\lambda(t) \leq \lambda_{max} = 5000 \forall t \in [0, 12]$. Then we can use the “thinning method” to simulate a non-homogeneous Poisson process with intensity $\lambda(t)$ by

1. Simulate an HPP (as described in 3e)) with intensity $\lambda_{max} = 5000$, giving simulated arrival times $s_1, s_2, \dots, s_N$.
2. Simulate N variables $u_1, \dots, u_N$ from the $U[0, 1]$ distribution.
3. For each simulated arrival time s_i , accept the time with probability $\lambda(s_i)/5000$, i.e. accept the time if $u_i \leq \lambda(s_i)/5000$, otherwise delete this time.
4. Return the remaining arrival times.

Problem 4

We wish to draw samples from a distribution with density $f(x)$ with

$$f(x) \propto \ln(x) * \exp\left(-\frac{(x-5)^2}{2\tau^2}\right), \quad x \geq 1, \tau > 0.$$

For this we are implementing two MCMC algorithm.

a)

```
gx <- function(x,tau=2){
  if(x<1) res <- 0
  else res <- log(x)*exp(-0.5 * (x-5)^2 / tau^2)
  return( res )
}

# Algorithm 1
MCMC.1 <- function(gx,tau=2,sigma=1,theta1=1,n.iter=10000){
  res <- numeric(n.iter)
  res[1] <- theta1
  wt.old <- gx(res[1],tau)/dnorm(res[1],mean=5,sd=sigma)
  for(i in 2:n.iter){
    thetaStar <- rnorm(1,mean=5,sd=sigma)
    wt.star <- gx(thetaStar,tau)/dnorm(thetaStar,mean=5,sd=sigma)
    alpha <- min(1.0,wt.star/wt.old)
    if(thetaStar>=1 & runif(1)<alpha){
      res[i] <- thetaStar
      wt.old <- wt.star
    } else {
      res[i] <- res[i-1]
    }
  }
}
```



```

    }
  }
  return(res)
}

```

What type of algorithm is the function MCMC.1? Explain and justify your answer!

The first algorithm is an Independent Metropolis Hasting algorithm.

The density kernel is defined by $g(x) = \ln(x) * \exp\left(-\frac{(x-5)^2}{2\tau^2}\right)$, $x \geq 1, \tau > 0$. First, we can recognize that $\exp\left(-\frac{(x-5)^2}{2\tau^2}\right)$ is the kernel of truncated normal distribution with mean $\mu = 5$ and variance τ^2 because $x \geq 1$.

The proposal mechanism `thetaStar <- rnorm(1,mean=5,sd=sigma)` is independent of the previous state, i.e. $\theta^* \sim q(\cdot|\theta_{t-1}) = r(\cdot)$. In our case, $r(\theta)$ is the probability density function of a normal distribution with mean $\mu = 5$ and variance σ^2 , i.e. $\theta^* \sim N(\mu = 5, \sigma^2)$. Then, we calculate the importance weight of the old state `wt.old <- gx(res[1],tau)/dnorm(res[1],mean=5,sd=sigma)`, i.e. $\frac{g(\theta_{t-1})}{r(\theta_{t-1})}$ and the importance weight of the proposal `wt.star <- gx(thetaStar,tau)/dnorm(thetaStar,mean=5,sd=sigma)`, i.e. $\frac{g(\theta^*)}{r(\theta^*)}$.

As $x \geq 1$ in the target distribution, we only accept if $\theta^* \geq 1$ and $U < \alpha$, with

$$\alpha = \min\left(1, \frac{\frac{g(\theta^*)}{r(\theta^*)}}{\frac{g(\theta_{t-1})}{r(\theta_{t-1})}}\right)$$

b)

```

# Algorithm 2
MCMC.2 <- function(gx,tau=1,sigma=1,theta1=2,n.iter=10000){
  res <- numeric(n.iter)
  res[1] <- theta1
  for(i in 2:n.iter){
    thetaStar <- rnorm(1,mean=res[i-1],sd=sigma)
    alpha <- min(1,gx(thetaStar,tau)/gx(res[i-1],tau))
    if(thetaStar>=1 & runif(1)<alpha){
      res[i] <- thetaStar
    } else {
      res[i] <- res[i-1]
    }
  }
  return(res)
}

```

What type of algorithm is the function MCMC.2? Explain and justify your answer!

The second algorithm is a Random Walk Metropolis Hasting (RWMH) algorithm.

The proposal mechanism `thetaStar <- rnorm(1,mean=res[i-1],sd=sigma)` depends on the previous state, i.e. $\theta^* \sim N(\theta_{t-1}, \sigma^2)$. The accept probability is given by `alpha <- min(1,gx(thetaStar,tau)/gx(res[i-1],tau))`, i.e.

$$\alpha = \min\left(1, \frac{g(\theta^*)}{g(\theta_{t-1})}\right)$$

.

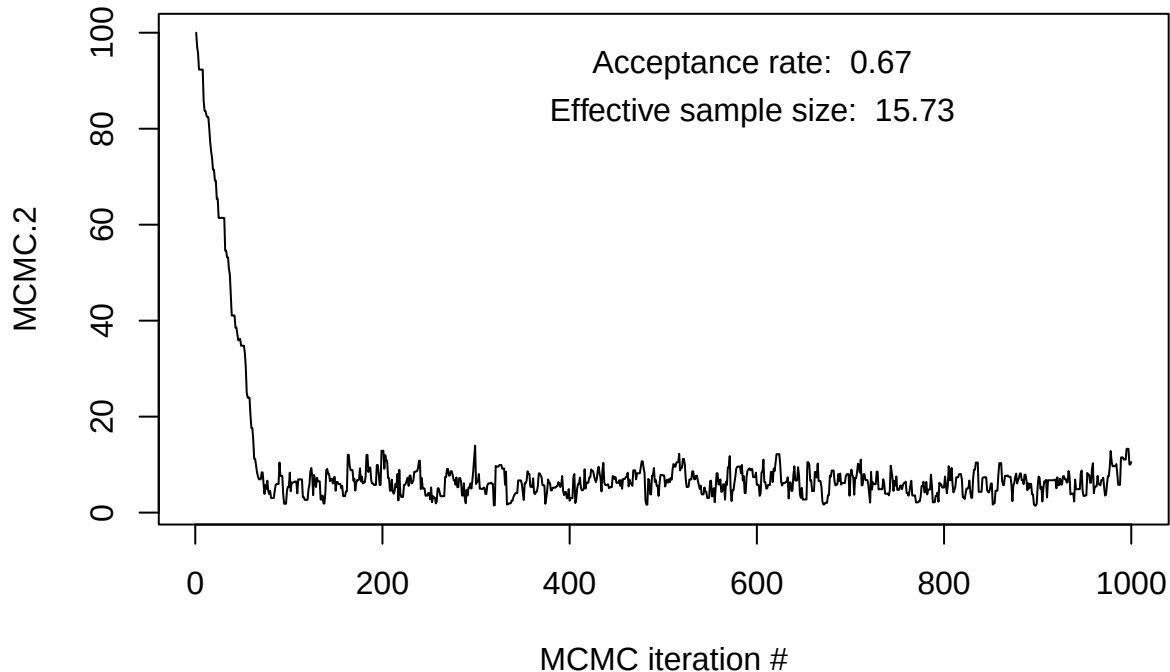
c)

Which of the algorithm would you prefer for the given goal? Explain you answer

The first algorithm, i.e. the independent Metropolis Hasting sampler, is to be preferred. It produces an uncorrelated sample of the target distribution. Further we can tune the algorithm in a way to have as high acceptance probability as possible. The RWMH algorithm produces a correlated sample and the algorithm should have between 20% – 40% acceptance probability. Thus, RWMH is less efficient.

d)

Below you see a trace plot of the MCMC chain generated by the second algorithm using the call `MCMC.2(gx,tau=3,theta1=100,sigma=3,n.iter=1000)` and two values to quantify the MCMC output. Can we use the generated sample as it is, or do you suggest to improve the generated sample for the target distribution? Why and what are your suggestions?



It is not recommended to use the generated sample for the target density $f(x)$ without modifications. First, we need either to adjust the initial value (which is far off from the stationary regime), or we need to delete the first 100 – 200 samples (burn in period). Otherwise, the first 100 – 200 samples of the MCMC will bias our statistical inferences. Further, we would like to select a different σ for the proposal in the RWMH algorithm in order to reach an acceptance probability between 20% – 40%. In the current setting we have a too high acceptance probability which indicates a too small σ . Both measures would lead to a larger effective sample size.